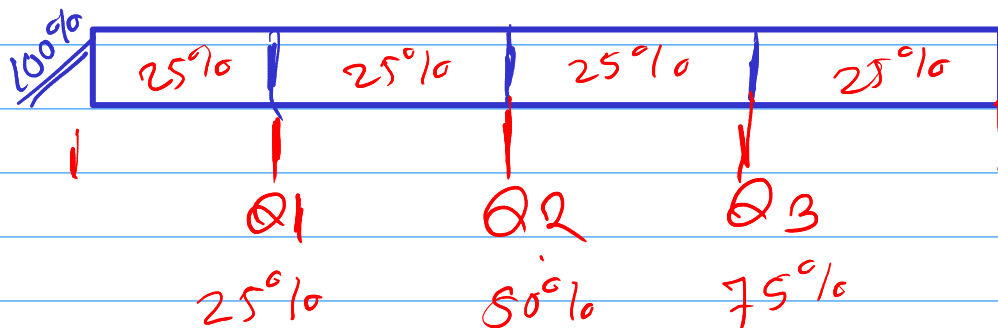


Outlier Detection

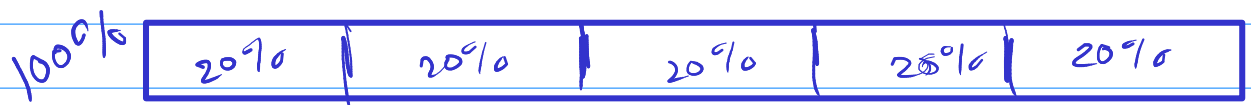
Quantile and Percentile

Quantile are statistical measure used to divide a set of numerical data into equal-sized groups, which groups containing an equal no. of Data or observation.

① Quartile $\Rightarrow$ Divide the data into four equal parts.



② Quintile $\Rightarrow$ Divide the data into five equal parts.



③ Decile % $\Rightarrow$ We divide the data into 10 equal parts.

$$10 \mid (0 \mid 0) \quad - \quad - \quad - \quad - \quad 1 \mid 10$$

④ Percentile $\Rightarrow$ We divide the data into 100 equal parts.

① Data should be sorted from low to high.

② you are finding the location of an observation
it is not the actual data point.

$$P_L = \frac{P}{100} (N+1)$$

PL $\rightarrow$ Location

$p \Rightarrow q$

$N = \text{Total no of Data Population}$

Males

Marks

x_1 x_2 x_3 x_4 x_5 x_6 x_7 x_8 x_9 x_{10}
= 78, 82, 84, 88, 91, 93, 94, 96, 98, 99

75% percentile

$$PL = \frac{P}{100} (N+1) = \frac{75}{100} (10+1) = \frac{3}{4} (11)$$

Location

$$= 3\frac{3}{4} = 8.25$$

$$= 8^{th} + 0.25 (9^{th} - 8^{th})$$

$$= 96 + \frac{(98 - 96) \times 0.25}{0.5}$$

$$= 96 +$$

$$= 96.5 \rightarrow \underline{\underline{75\% Rank}}$$

Outlier Detection Technique

5-Number Summary

→ It is a descriptive statistics that provides a summary of a dataset. It consists of five values that divide the data into four equal parts also known as Quartile.

(The fine no. includes :-

- ① Minimum Value i.e. Lower Boundry.
 (The minimum value in the dataset is not the lower boundry).

[7, 8, 9, 10, 15, 20, 04, 33, 19, 20]

$$\text{Lower boundry} = Q_1 - 1.5 (IQR)$$

- ② First Quartile (Q_1) $\Rightarrow$ It is the value in the dataset at 25% Rank(location)

- ③ Second Quartile (Q_2) $\Rightarrow$ It is the value in the dataset at 50% Location.
 It may be median.

- ④ Q_3 (Third Quartile) $\Rightarrow$ It is the value in the dataset at 75% Location.

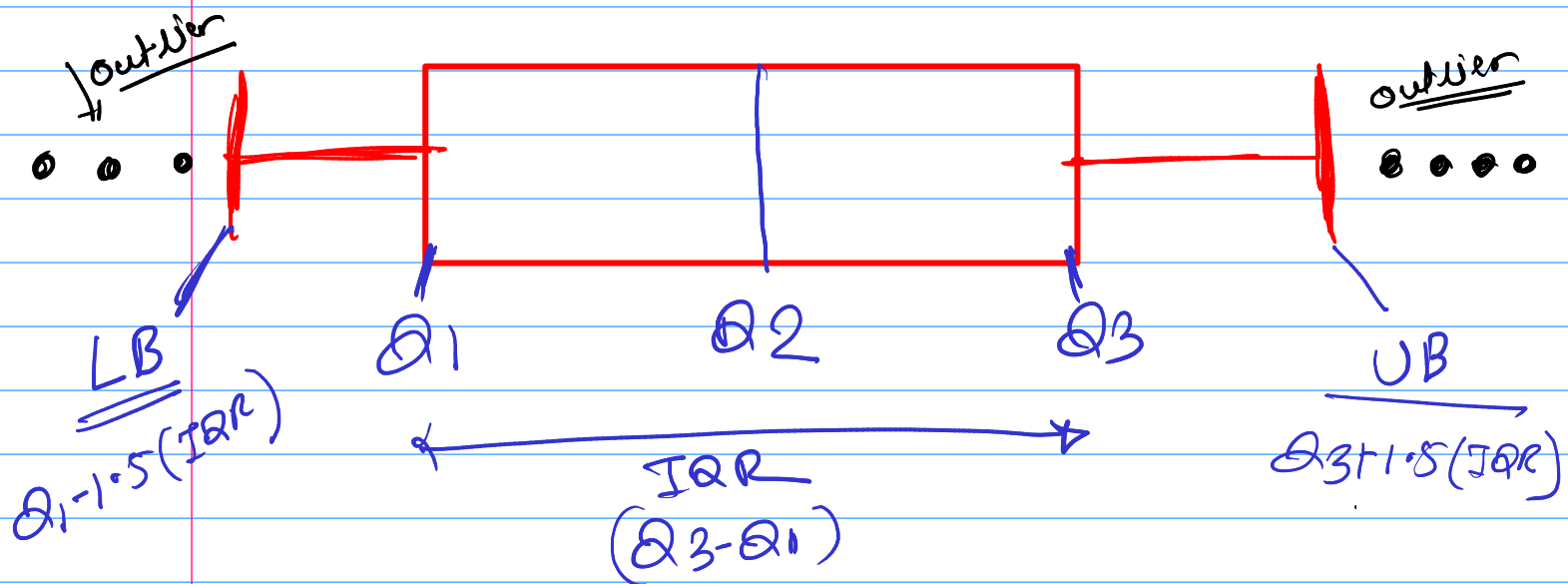
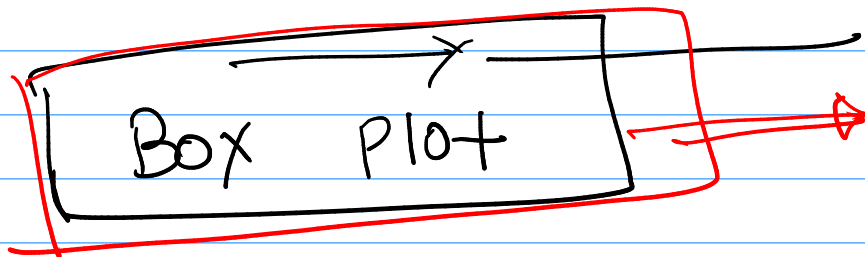
⑤ Upper Boundary $\Rightarrow$ It is not the max value in the dataset

$$UB = Q_3 + 1.5(IQR)$$

IQR $\Rightarrow$ Inter Quartile Range:-

$\Rightarrow$ The difference between third Quartile & the first Quartile.

$$IQR = Q_3 - Q_1$$



data : $\rightarrow$ 17, 18, 19, 21, 23, 24, 27, 28, 299, 311 —

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 x_1 x_2 x_3 x_4 x_5 x_6 x_7 x_8 x_9 x_{10}

$$Q_1 = \frac{P}{100}(N+1) = \frac{25}{100}(10+1) = \frac{1}{4}(11) = \frac{11}{4} = \underline{\underline{2.75}}$$

Location

$$= 2^{\text{nd}} + 0.75(3^{\text{rd}} - 2^{\text{nd}})$$

$$= 18 + 0.75(19 - 18) = \boxed{18.75}$$

$$Q_2 = \frac{P}{100}(N+1) = \frac{50}{100}(11) = \frac{11}{2} = \underline{\underline{5.5}}$$

$$= \frac{5^{\text{th}} + 6^{\text{th}}}{2} = \frac{24 + 23}{2} = \boxed{23.5}$$

$$Q_3 = \frac{P}{100}(N+1) = \frac{75}{100}(11) = \frac{33}{4} = \underline{\underline{8.25}}$$

$$= 8^{\text{th}} + 0.25(9^{\text{th}} - 8^{\text{th}})$$

$$= 28 + 0.25(299 - 28)$$

$$= \boxed{95.75} = Q_3$$

$$IQR = Q_3 - Q_1 = 95.75 - 18.75$$

$$\boxed{IQR = 77}$$

$$LB = Q_1 - 1.5(IQR)$$

$$= 18.75 - 1.5(77)$$

$$\underline{\underline{115.5}}$$

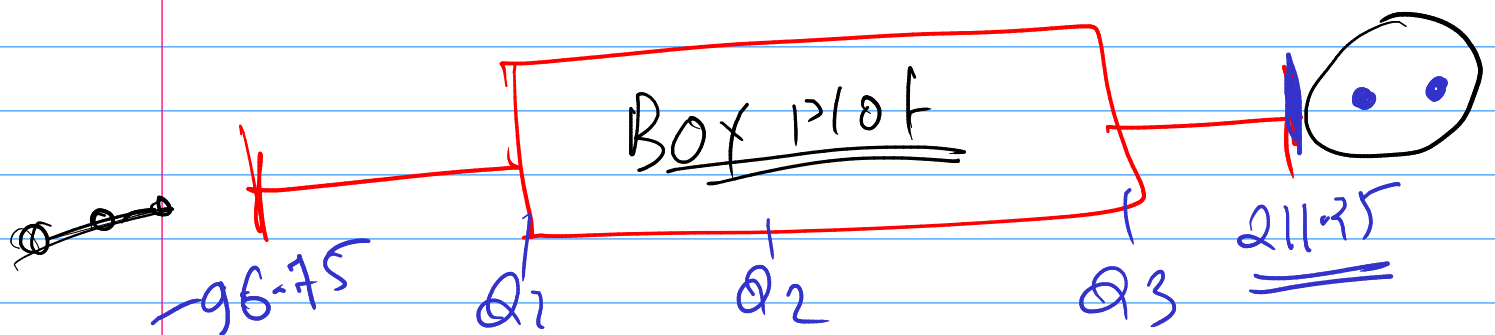
$$\boxed{-96.75}$$

$$UB = Q_3 + 1.5(IQR)$$

$$= 95.75 + 1.5(77)$$

$$\{(\infty)\}$$

$$\boxed{211.25} \quad \boxed{UB}$$



Box plot

